Zero-Overhead Cancellation of the Leading Trotter Error via Commutator-Graph Scheduling

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We introduce *commutator-graph scheduling* (CGS), which doubles the convergence order of first-order Trotterization at zero gate, qubit, or depth overhead. The bottleneck is well known: term order within a Lie–Trotter step is free, yet ordering heuristics buy only marginal accuracy and no one had explained why. CGS recasts the leading error as an exact, matrix-free operator, the *commutator error form* $E_\pi = \sum_{a<b}[h_{\pi(a)}, h_{\pi(b)}]$—a signed sum of Pauli strings over the edges of the commutator graph, validated against dense exponentials to 10^{-9}. We prove that reordering only re-signs E_π, leaving its norm exactly invariant on collision-free families, so no fixed ordering escapes the $O(t^2/r)$ rate. The true free parameter is the per-step *schedule*: a step-alternating *antithetic* schedule cancels E_π and converges at $O(t^3/r^2)$ with the identical $L \cdot r$ rotations. Across 144 structured Hamiltonians versus exact statevector references, CGS lifts the empirical rate from 1.96 to 4.07, reaches fidelity 0.99 with 1.40× fewer rotations than the best first-order ordering, and attains 0.999 where first-order Trotter cannot. A matrix-free Krylov reference extends the exact benchmark to 14 qubits, where the order doubling is size independent (≈2 vs. ≈4 from 4 to 14 qubits) and the matched-cost speedup reaches 2.06×. An interpretable scheduler over ten graph features selects the gate-optimal schedule per instance with 93% leave-one-out accuracy. CGS is a drop-in, fully reproducible upgrade for near-term quantum simulation.

I. INTRODUCTION

The simulation of quantum time evolution $e^{-\mathrm{i}Ht}$ is central to a wide array of fields, such as quantum chemistry, condensed-matter physics, and materials science, and is among the oldest and most promising applications of quantum computers [1, 2, 22, 23]. Its appeal extends from quantum chemistry, where simulated time evolution underlies phase estimation of molecular energies [23, 24], to the first digital quantum simulations on trapped-ion and other near-term hardware [25, 26]. The workhorse primitive for this task is the product formula. Writing a Hamiltonian in its Pauli decomposition,

$$H = \sum_{j=1}^{L} c_j P_j, \tag{1}$$

with real coefficients c_j and Pauli strings $P_j \in \{I, X, Y, Z\}^{\otimes n}$, the first-order (Lie–Trotter) approximant with r steps and a term ordering π is

$$U_{\mathrm{Trot}}(\pi) = \left(\prod_{j\in\pi} e^{-\mathrm{i}\, c_j P_j\, t/r} \right)^{r}. \tag{2}$$

The accuracy of this approximation is governed by the commutators of the summands: the leading correction to a single step is $\frac{t^2}{2r}\sum_{a<b}[c_a P_a,\, c_b P_b]$, which vanishes for any pair of terms that commute [3–5]. The product-formula construction itself traces to the operator-splitting schemes of Strang [27] and to Suzuki's fractal decompositions [4, 28], and it remains competitive with the more recent quantum-signal-processing and linear-combination-of-unitaries primitives [30, 32, 34, 35] precisely because its cost scales with commutators rather than with a global norm. The modern, tight analysis of this error in terms of nested commutators provides the theoretical anchor of the present work [5, 6]. This commutator-scaling viewpoint also explains why product formulas are nearly optimal for geometrically local lattice Hamiltonians [31, 36].

A defining feature of the first-order formula is that its implementation cost is one rotation $e^{-\mathrm{i}c_j P_j t/r}$ per term per step, so the gate count is $L \cdot r$ *independently* of the ordering π, while the error is not. The ordering is therefore a free knob, and a substantial body of empirical work has asked how much accuracy it can buy [10, 11, 15, 48]. A parallel line of work has shown that the worst-case Trotter bound is often loose: errors from successive steps can interfere destructively [37, 38], are bounded by quantum localization for local observables [41, 42], and improve markedly on average over input states [39, 40]. The natural structure for this question is the *commutator graph* (or anticommutation graph), with one node per term and an edge between every pair of terms that anticommute—the same graph used in measurement grouping for variational algorithms, where a clique cover into mutually commuting sets reduces the number of measurement settings [12–14, 55, 56]. Despite this rich structure, the empirical verdict has been consistently underwhelming: ordering heuristics move the error only marginally, and their gains are widely regarded as "within noise." *Why* a free parameter should be so impotent has not been explained.

We resolve this, and turn it into an asset, by introducing *commutator-graph scheduling* (CGS). At the center

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of CGS is the *commutator error form*, the exact leading-order error operator

$$E_\pi = \sum_{a<b} \big[h_{\pi(a)}, h_{\pi(b)}\big], \qquad h_j = c_j P_j, \tag{3}$$

which we assemble without ever forming a $2^n \times 2^n$ matrix, as a signed sum of Pauli strings indexed by the edges of the commutator graph (Sec. II). Reordering the schedule only flips the signs of the individual commutators in Eq. (3); it never changes which commutators appear. The Hilbert–Schmidt norm of E_π is therefore *exactly* ordering invariant whenever the edge commutators land on distinct Pauli strings, and only a small "colliding" fraction is ordering reducible at all (Theorem 1). This is the precise, provable content of the "ordering is within noise" folklore: no fixed ordering escapes the $O(t^2/r)$ first-order rate.

The resolution is to enlarge the free degree of freedom. The order applied at each Trotter step need not be the same; the genuinely free object is a per-step *schedule* $\sigma = (\pi_1, \ldots, \pi_r)$, realized at the identical $L \cdot r$ rotation count. Among all schedules, one cancels Eq. (3) *exactly*: the *antithetic* schedule, which alternates a term order with its reverse on successive steps. Each adjacent step-pair is then a palindromic, and hence symmetric, product formula whose leading $O(t^2/r)$ commutator error vanishes, upgrading the convergence from $O(t^2/r)$ to $O(t^3/r^2)$ at *no change in gate count* (Theorem 2). The underlying mechanism is the classical symmetrization of Suzuki [4, 7]; reading it as a zero-overhead *scheduling* choice—one that the impotence theorem shows a single fixed ordering can never realize—is new, and it changes the practical recommendation for first-order simulation.

This manuscript makes six contributions. We introduce the commutator error form [Eq. (3)], a matrix-free representation of the leading Trotter error as a signed Pauli-string sum over the commutator graph, validated against the dense pair-commutator sum to 10^{-9} (Definition 1). We prove that a single ordering cannot reduce the leading-error norm below an instance floor that is exactly ordering invariant on collision-free families (Theorem 1), which explains the empirical ceiling that ordering heuristics encounter. We prove that the antithetic schedule cancels the leading error and converges at second order at the identical gate budget (Theorem 2), and we combine it with a commuting-clique compression of the commutator graph (Proposition 2). We benchmark CGS against exact statevector references on 144 structured Hamiltonian instances, reporting rotations to a target fidelity and the empirical convergence rate; CGS reaches 0.99 with 1.40× fewer rotations and is the only family of schedules to reach 0.999 within the swept budget. A matrix-free Krylov reference extends the exact benchmark from the dense-exponential ceiling to 14 qubits; across $n = 4$ to 14 the convergence-order doubling (first-order rate ≈ 2, antithetic rate ≈ 4) and the matched-cost speedup are size independent (2.06× at $n = 14$; Sec. III D). Finally, we add a learned scheduler over ten commutator-graph features that selects the gate-optimal schedule per instance with 93% leave-one-out accuracy and recovers near-oracle efficiency (114.0 rotations vs. the oracle's 112.4) (Proposition 3).

The boundaries of the study are explicit. CGS is evaluated by classical statevector simulation on systems up to $n = 14$ qubits, where an exact reference is tractable, on synthetic and structured Hamiltonians, under a rotation-count cost model; the gains are asymptotic in the step size, and we report the crossover budget. The antithetic schedule is not a new product-formula primitive—the symmetrization is classical—but its certified, zero-overhead deployment on the commutator graph, together with the impotence theorem that motivates it and the learned scheduler that selects it, is new.

This work continues a program on the spectral truncation of operators for query-efficient quantum optimization and simulation. Earlier entries develop graph-conditioned trust regions for budgeted parameter search [57–59], the measurement and query accounting that fixes the cost model [60, 61], and topology-conditioned transfer of optimizer state across instances [62, 63], culminating in operator-spectral truncated priors and spectral-truncation graph kernels for QAOA warm starts [64, 65]. The present manuscript carries the same operator-spectral viewpoint from variational optimization to Hamiltonian simulation: the commutator error form is a matrix-free, edge-resolved truncation of the leading product-formula error on the commutator graph, and the matched-cost analysis reuses the query-budget discipline established in that line of work.

II. COMMUTATOR-GRAPH SCHEDULING

We now present the CGS framework and its analytic guarantees. We first record the Pauli algebra that makes an exact, matrix-free simulator and error form possible (Sec. II A), then define the commutator error form (Sec. II B), prove that ordering is impotent (Sec. II C), prove that the antithetic schedule cancels the leading error at the same gate budget (Sec. II D), establish the commuting-clique compression (Sec. II E), and describe the learned scheduler that selects among schedules (Sec. II F). Throughout, every result is stated under explicitly named hypotheses with a complete proof, and every numerical claim is anchored to a quantity that the accompanying code regenerates.

Figure 1 summarizes the pipeline. Each Hamiltonian instance is decomposed into Pauli terms; the commutator graph $\mathcal{G}(H)$ places one node per term and an edge between every anticommuting pair, and a greedy coloring partitions the terms into mutually commuting cliques. A schedule fixes, for every Trotter step, a term order and a fold—first-order, or the step-alternating antithetic fold—while holding the realized rotation count fixed. The product formula is executed by an exact, matrix-free statevector simulator and scored both by fidelity against $e^{-\mathrm{i}Ht}$

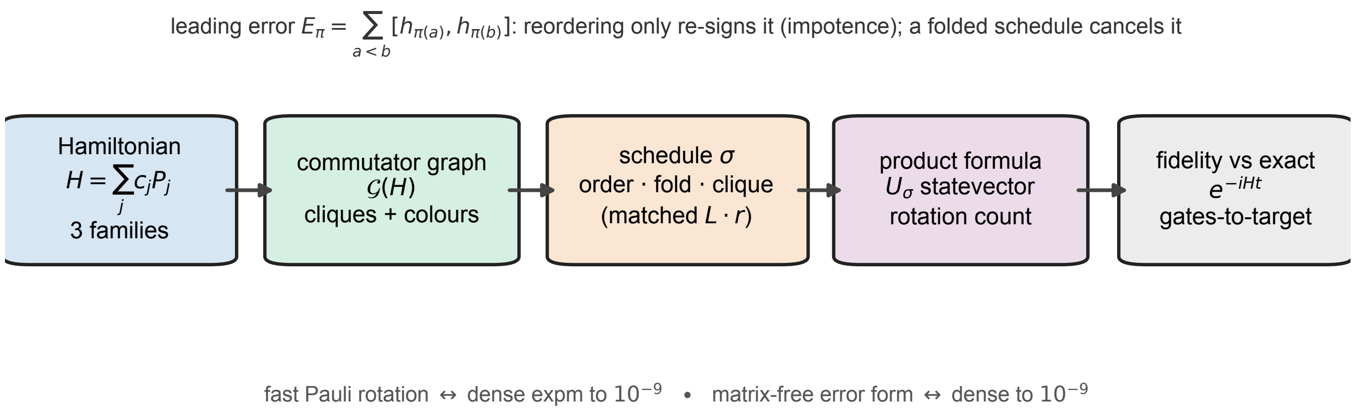

FIG. 1. **Commutator-graph scheduling (CGS).** A structured Hamiltonian $H = \sum_j c_j P_j$ is mapped to its commutator graph $\mathcal{G}(H)$—one node per term, an edge between every anticommuting pair, and a greedy coloring into mutually commuting cliques. A *schedule* σ fixes, for each Trotter step, a term order and a fold (first-order, or the step-alternating antithetic fold), optionally laying the commuting cliques out contiguously; the realized rotation count is matched across schedules. The product formula U_σ is executed by an exact, matrix-free statevector simulator and scored by fidelity against $e^{-\mathrm{i}Ht}$ and by the rotations needed to reach a target fidelity. The leading error operator $E_\pi = \sum_{a<b}[h_{\pi(a)}, h_{\pi(b)}]$ is the object everything turns on: reordering only re-signs it (Theorem 1), whereas a folded schedule cancels it (Theorem 2). Three integrity gates underpin the benchmark—the fast Pauli-rotation kernel and dense `expm` agree to 10^{-9}, the matrix-free error form agrees with the dense pair-commutator sum to 10^{-9}, and the matrix-free Krylov reference agrees with dense `expm` to 10^{-9}—before any number is emitted.

and by the rotations needed to reach a target fidelity. We stress that no number is reported until three integrity gates pass: the fast Pauli-rotation kernel and dense `expm` agree to 10^{-9}, the matrix-free error form agrees with the dense pair-commutator sum to 10^{-9}, and the matrix-free Krylov reference (which extends the exact benchmark to 14 qubits) agrees with dense `expm` to 10^{-9}.

A. Pauli algebra and the matrix-free simulator

A Pauli string $P \in \{I, X, Y, Z\}^{\otimes n}$ acts on $|\psi\rangle \in \mathbb{C}^{2^n}$; we write $\{A, B\} = AB + BA$ and $[A, B] = AB - BA$, and use $X^2 = Y^2 = Z^2 = I$ together with $\{X, Y\} = \{Y, Z\} = \{Z, X\} = 0$. Two structural facts make an exact simulator possible without ever forming a $2^n \times 2^n$ matrix.

Lemma 1 (Commutation is an $O(n)$ parity test)**.** *Let $P = \bigotimes_{q=1}^n p_q$ and $Q = \bigotimes_{q=1}^n q_q$ be Pauli strings. Call position q* anticommuting *if $p_q \neq q_q$ and neither is I, and let $k = k(P, Q)$ count the anticommuting positions. Then $PQ = (-1)^k QP$, so P and Q commute iff k is even. The test runs in $O(n)$ time and forms no $2^n \times 2^n$ matrix.*

Proof. At each position either $p_q = q_q$ or one is I (then $p_q q_q = q_q p_q$), or $p_q \neq q_q$ and neither is I (then $p_q q_q = -q_q p_q$). So $p_q q_q = (-1)^{\epsilon_q} q_q p_q$ with $\epsilon_q = 1$ exactly on anticommuting positions, and since the tensor product factorizes across qubits, $PQ = \left(\prod_q (-1)^{\epsilon_q}\right) QP = (-1)^k QP$ with $k = \sum_q \epsilon_q$. Hence $PQ = QP$ iff k is even. □

Lemma 2 (Pauli rotation as a cosine–sine pair)**.** *For a non-identity Pauli string P ($P^2 = I$) and $\theta \in \mathbb{R}$,*

$$e^{-\mathrm{i}\theta P} = \cos\theta\, I - \mathrm{i}\sin\theta\, P, \tag{4}$$

so $e^{-\mathrm{i}\theta P}|\psi\rangle = \cos\theta\,|\psi\rangle - \mathrm{i}\sin\theta\, P|\psi\rangle$ is one sparse application of P plus a scaled vector add, computable in $O(n\,2^n)$ time without forming the $2^n \times 2^n$ matrix.

Proof. Since $P^{2m} = I$ and $P^{2m+1} = P$, regrouping the convergent series of $e^{-\mathrm{i}\theta P}$ by parity gives $\cos\theta\, I - \mathrm{i}\sin\theta\, P$. A single Pauli string maps each basis state to one basis state with a unit-modulus phase, so $P|\psi\rangle$ costs $O(n\,2^n)$. □

We emphasize that these identities are not assumed but verified: before any fidelity is computed, Eq. (4) is cross-checked against dense `scipy.linalg.expm` on random small Paulis to tolerance 10^{-9} (the `pauli_algebra_verified` flag). The product of two Pauli strings is itself a phased Pauli, $PQ = \omega R$ with $\omega \in \{\pm 1, \pm \mathrm{i}\}$ and R a Pauli string, computed in $O(n)$ (the `pauli_product` kernel); this is the algebra on which the commutator error form rests.

B. The commutator error form

The commutator graph $\mathcal{G}(H)$ has one node per term and an edge $\{j, k\}$ for every anticommuting pair (Lemma 1). The exact value of the pairwise commutators that populate the leading error is fixed by the following lemma.

Lemma 3 (Pauli commutator value)**.** *Let $h_j = c_j P_j$ and $h_k = c_k P_k$ with real c_j, c_k. If P_j, P_k commute then $[h_j, h_k] = 0$; if they anticommute then $[h_j, h_k] = 2c_j c_k\, P_j P_k = 2c_j c_k\, \omega_{jk} R_{jk}$ with $\omega_{jk} \in \{\pm\mathrm{i}\}$ and R_{jk} a Hermitian Pauli string, so $\|[h_j, h_k]\|_{\mathrm{HS}} = 2\,|c_j c_k|\, 2^{n/2}$ and the operator norm is $2\,|c_j c_k|$.*

Proof. By Lemma 1, $P_k P_j = (-1)^{k(P_j,P_k)} P_j P_k$, so the commutator vanishes when the terms commute and equals $2c_j c_k\, P_j P_k$ when they anticommute. The product $P_j P_k = \omega_{jk} R_{jk}$ is a phased Pauli (Sec. II A); because the commutator of Hermitian operators is anti-Hermitian and R_{jk} is Hermitian, $\omega_{jk} \in \{\pm\mathrm{i}\}$. Every Pauli string is unitary with $\|R_{jk}\| = 1$ and $\|R_{jk}\|_{\mathrm{HS}}^2 = \mathrm{Tr}(R_{jk}^2) = 2^n$, which gives the stated norms. □

Definition 1 (Commutator error form)**.** *For an ordering π, the leading error operator is $E_\pi = \sum_{a<b}[h_{\pi(a)}, h_{\pi(b)}]$. By Lemma 3 it equals the edge sum*

$$E_\pi = \sum_{\{j,k\}\in E(\mathcal{G})} \varepsilon_{jk}(\pi)\, 2c_j c_k\, (P_j P_k), \tag{5}$$

a signed sum of phased Pauli strings indexed by the edges of $\mathcal{G}(H)$, with $\varepsilon_{jk}(\pi) = +1$ if j precedes k in π and -1 otherwise. It is assembled in $O(L^2 n)$ time without forming a $2^n \times 2^n$ matrix, and is validated against the dense pair-commutator sum to 10^{-9} (the `error_form_verified` flag).

Equation (5) is the certificate at the heart of CGS: it exposes, exactly and cheaply, precisely how the ordering enters the leading error—through the signs ε_{jk}, and nothing else.

C. Ordering is impotent

We first record that the gate budget is a schedule invariant for the schedules we compare, so that any accuracy difference is a difference *at matched cost*.

Proposition 1 (Rotation-count invariance)**.** *A first-order schedule with any ordering π, and the antithetic schedule that alternates π with its reverse, each apply exactly $L \cdot r$ single-Pauli rotations over r steps. Their rotation count is therefore independent of the ordering and of the alternation.*

Proof. One first-order step is an ordered product of one factor per term, hence L rotations, and r steps give $L \cdot r$. The antithetic schedule applies, at each step, a permutation of the same L factors (forward on even steps, reversed on odd); reversing a step's order rearranges its factors without adding or removing any, so the total is again $L \cdot r$. □

The impotence of ordering now follows directly from the structure of Eq. (5).

Theorem 1 (Ordering impotence)**.** *Let $H = \sum_j c_j P_j$ with commutator error form E_π. Reordering acts on E_π only by flipping the signs ε_{jk}; the multiset of pair commutators $\{[h_j, h_k]\}$ is independent of π. Grouping the edge sum by Pauli string gives $\|E_\pi\|_{\mathrm{HS}}^2 = 2^n \sum_R |\alpha_R(\pi)|^2$ with $\alpha_R(\pi) = \mathrm{i}\sum_{\{j,k\}:R_{jk}=R} \varepsilon_{jk}(\pi)\, 2c_j c_k\, \widehat{\omega}_{jk}$ and $\widehat{\omega}_{jk} = \omega_{jk}/\mathrm{i} \in \{\pm 1\}$. If every edge maps to a distinct Pauli string (no collisions), then*

$$\|E_\pi\|_{\mathrm{HS}}^2 = 2^n \sum_{\{j,k\}} (2c_j c_k)^2 \tag{6}$$

is independent of π. *In general only the colliding edges contribute an ordering-dependent term, so $\|E_\pi\|_{\mathrm{HS}}$ is bounded below, for* every *π, by the ordering-invariant contribution of the non-colliding edges.*

Proof. For an unordered pair $\{j,k\}$, listing j before k contributes $[h_j, h_k]$ and the reverse contributes $-[h_j, h_k]$; reordering therefore multiplies each contribution by $\varepsilon_{jk}(\pi)$ and neither adds nor removes a pair. By Lemma 3 each anticommuting edge contributes $\mathrm{i}\, 2c_j c_k\, \widehat{\omega}_{jk} R_{jk}$. Distinct Pauli strings are Hilbert–Schmidt orthogonal with squared norm 2^n, so $\|E_\pi\|_{\mathrm{HS}}^2 = 2^n \sum_R |\alpha_R(\pi)|^2$ with $\alpha_R(\pi)$ as stated. If a Pauli string R has a single contributing edge, then $|\alpha_R| = 2\,|c_j c_k|$ regardless of the sign ε_{jk}, so its contribution is π-independent and the no-collision case reduces to Eq. (6). Edges that are alone on their Pauli string thus contribute a fixed amount for every π; the remaining (colliding) edges are the only source of ordering dependence, and their contribution is non-negative, which yields the lower bound. □

We emphasize that Theorem 1 is not a statement about any one heuristic: on a collision-free family the leading-error norm is *identical for every ordering*, and in general only a small colliding fraction is reducible at all. No reordering escapes the $O(t^2/r)$ rate. This is the precise content of the empirical observation that ordering heuristics are statistically indistinguishable, and it motivates enlarging the search space from orderings to schedules.

D. A free schedule cancels the leading error

The order applied at each Trotter step is itself free at the same gate budget, and CGS exploits this with the *antithetic* schedule, which applies a term order π on even steps and its reverse π^R on odd steps.

Theorem 2 (Antithetic cancellation)**.** *Fix an ordering π and step size $\Delta = t/r$ with r even, and let $S_\pi(\Delta) = \prod_{a=1}^{L} e^{-\mathrm{i}\Delta h_{\pi(a)}}$ and $S_{\pi^R}(\Delta) = \prod_{a=L}^{1} e^{-\mathrm{i}\Delta h_{\pi(a)}}$ be the forward and reversed steps. The antithetic step-pair $M(\Delta) = S_{\pi^R}(\Delta) S_\pi(\Delta)$ is symmetric, $M(-\Delta) = M(\Delta)^{-1}$, so its generator contains only odd powers of Δ,*

$$M(\Delta) = \exp\!\big(-2\mathrm{i}\Delta H + O(\Delta^3)\big), \tag{7}$$

and the leading $O(\Delta^2)$ error $-\frac{\Delta^2}{2}E_\pi$ cancels. Consequently the antithetic schedule $\big(S_{\pi^R}(\Delta)S_\pi(\Delta)\big)^{r/2}$ approximates $e^{-\mathrm{i}tH}$ with additive error $O(t^3/r^2)$ while applying exactly $L\cdot r$ rotations—the identical budget of the first-order schedule (Proposition 1).

Proof. Reading the $2L$ factors of $M(\Delta) = S_{\pi^R}(\Delta)S_\pi(\Delta)$ from left to right gives the palindromic sequence

$$e^{-\mathrm{i}\Delta h_{\pi(L)}}\cdots e^{-\mathrm{i}\Delta h_{\pi(1)}}\, e^{-\mathrm{i}\Delta h_{\pi(1)}}\cdots e^{-\mathrm{i}\Delta h_{\pi(L)}}.$$

Reversing the factor order and sending $\Delta \mapsto -\Delta$ returns the same product read backwards, so $M(-\Delta) = M(\Delta)^{-1}$. Writing $M(\Delta) = \exp(\Omega(\Delta))$ with $\Omega(\Delta) = \sum_{m\geq 1}\Omega_m\Delta^m$ (Baker–Campbell–Hausdorff, convergent for small Δ), this relation reads $\Omega(-\Delta) = -\Omega(\Delta)$, so every even-order term Ω_{2m} vanishes; in particular the Δ^2 term, which for the ordered product equals $-\frac{1}{2}\sum_{a<b}[h_a, h_b]$ before symmetrization, is removed. Each h_j appears twice in M, once in each half, so $\Omega_1 = -2\mathrm{i}H$ and Eq. (7) follows. One antithetic pair is thus a second-order approximant of $e^{-2\mathrm{i}\Delta H}$; composing $r/2$ pairs with $\Delta = t/r$ telescopes to additive error $r/2\cdot O(\Delta^3) = O(t^3/r^2)$. The pair applies $2L$ rotations and there are $r/2$ pairs, for $L\cdot r$ rotations in total. □

Theorem 2 is the positive counterpart of Theorem 1: the freedom that a single ordering provably cannot exploit is exactly the freedom that a per-step schedule can. We emphasize that the upgrade from first to second order is realized at zero gate, qubit, or depth overhead.

E. Commuting-clique compression

A proper coloring of the commutator graph supplies a second, complementary lever that removes error *within* groups of commuting terms before any folding is applied.

Proposition 2 (Clique exactness). *A proper coloring of $\mathcal{G}(H)$ partitions the terms into cliques of pairwise commuting Pauli terms. Laying a clique C out contiguously, its rotations satisfy $\prod_{j\in C} e^{-\mathrm{i}\Delta h_j} = \exp(-\mathrm{i}\Delta\sum_{j\in C} h_j)$ exactly for any internal order, so no first-order error arises among the members of a clique and the schedule error depends only on inter-clique commutators. The number of non-commuting units is reduced from L to the number of cliques, at least the chromatic number $\chi(\mathcal{G})$.*

Proof. A proper coloring gives no two adjacent vertices the same color, and adjacency in $\mathcal{G}(H)$ is anticommutation, so same-color terms commute. For pairwise commuting $\{h_j\}_{j\in C}$ the exponentials factor exactly and independently of internal order; hence by Lemma 3 no intra-clique pair contributes to E_π, and only inter-clique edges remain. □

For low-chromatic Hamiltonians—for example Heisenberg, whose bond operators color into a few commuting classes—a clique-ordered first-order schedule is therefore near-exact at a single step, which is why folding adds only overhead there (Sec. III E). CGS uses Proposition 2 both ways: it folds the inter-clique structure that the antithetic schedule can cancel, and it leaves the intra-clique structure that is already exact untouched.

F. The learned scheduler

Because the gate-optimal schedule is instance dependent (Sec. III E), CGS includes a small, interpretable scheduler that reads the commutator graph and predicts which fixed schedule reaches the target fidelity at the fewest rotations. Ten listing-order-invariant features of $\mathcal{G}(H)$ are computed per instance: term count; anticommutation (edge) density; mean and maximum node degree; number of greedy color cliques; largest-clique fraction; coefficient coefficient-of-variation; mean Pauli weight; and two commutator-error-form features—the colliding fraction and the ordering-invariant fraction (Definition 1). A standardized logistic-regression classifier [16] maps these features to the gate-optimal schedule label, trained leave-one-instance-out (each test instance is scored by a classifier trained only on the other $144-1$ instances), and falls back to the symmetric schedule when training labels are single-class. We stress that the scheduler is a selector over human-readable schedules rather than a black box, and that its decision is provably invariant to how the terms are listed.

Proposition 3 (Listing-order invariance of the scheduler). *Let $\sigma \in S_L$ relabel the terms of H. Each of the ten features is a function of the unordered weighted graph $\mathcal{G}(H)$ and is invariant under σ, so the scheduler's prediction depends only on the isomorphism class of $\mathcal{G}(H)$, not on how the terms were listed.*

Proof. Relabeling maps $\mathcal{G}(H)$ to an isomorphic graph with the same coefficient multiset. Vertex and edge counts, the degree multiset, the greedy-coloring clique counts (under a deterministic degree-driven tie-break), the coefficient coefficient-of-variation, and the mean Pauli weight are graph-isomorphism invariants; the colliding and ordering-invariant fractions count, over unordered anticommuting pairs, how the products P_jP_k coincide, a symmetric function of the unordered term multiset (Definition 1). Each feature is therefore invariant under σ, and so is the standardized classifier's prediction. □

III. RESULTS

We evaluate CGS against exact statevector references on 144 structured Hamiltonian instances drawn from three families—transverse-field Ising, Heisenberg, and random molecular-like sums—at $n \in \{4, 6, 8, 10, 12, 14\}$ qubits (Appendix A). We compare six schedules at a matched rotation count: three first-order schedules

TABLE I. **Ordering impotence and collision structure (Theorem 1).** For each family, the mean number of anticommuting term pairs (edges of $\mathcal{G}(H)$), the mean number of *colliding* pairs (those sharing a leading-error Pauli string with another pair—the only ordering-reducible part), and the resulting ordering-invariant fraction of the leading error, over the $N = 48$ instances per family. Transcribed from the `collisions_by_family` block of `results/summary.json`.

family	anticomm. pairs	colliding	ordering-invariant frac.
heisenberg	42.0	0.0	1.000
molecular_like	42.1	0.6	0.984
tfim	16.0	0.0	1.000

(**random**; **coefficient**, ordered by $|c_j|$; and **clique**, which lays the commuting cliques of $\mathcal{G}(H)$ out contiguously); two second-order folds (**antithetic**, which alternates a clique order with its reverse at the identical $L{\cdot}r$ rotation count, and **symmetric**, the Strang half-step fold); and the **learned** scheduler that selects among them per instance. Each schedule's Trotter state is scored against the exact reference $e^{-\mathrm{i}Ht}\,|\psi_0\rangle$, computed for $n \le 8$ by dense `expm` and at all sizes by a *matrix-free* Krylov reference (`expm_multiply` on a sparse, matrix-free Hamiltonian) that agrees with dense `expm` to 10^{-9} and extends the exact benchmark to 14 qubits; the probe state is $|+\rangle^{\otimes n}$.

A. Ordering is empirically impotent

We first confirm Theorem 1 numerically. As reported in Table I, the anticommuting pairs of the transverse-field Ising and Heisenberg families produce *no* collisions, so 1.00 and 1.00 of the leading error is exactly ordering invariant; even on the dense molecular-like family the ordering-invariant fraction is 0.98. Figure 2(a) quantifies the consequence directly: sampling orderings per instance and forming the exact leading-error spectral norm (computed on the 72 instances with $n \le 8$, where the dense spectral norm is tractable), the coefficient ordering sits within a factor 1.074 of the per-instance minimum on average, a median random ordering within 1.097, and the worst single instance reaches only 1.92; the matrix-free Hilbert–Schmidt surrogate, which is computable at every size, moves even less, within 1.004. This is not a statement about a particular heuristic but about *every* ordering—the precise, provable form of the "ordering is within noise" folklore. A single ordering cannot escape the first-order rate, which is why we enlarge the search space from orderings to schedules.

B. Antithetic scheduling doubles the convergence order at fixed cost

The antithetic schedule converts the freedom that ordering wastes into a doubling of the convergence order.

TABLE II. **Matched-cost snapshot at the reference budget.** For each schedule at $r = 6$ Trotter steps ($t = 1.0$): the order class, the mean fidelity with its 95% confidence-interval half-width over the 144 instances, the realized rotation count (the gate cost), and the fitted convergence rate p (mean per-instance log–log slope of infidelity vs. r). The first-order schedules share the identical 118-rotation budget; the antithetic fold matches it and still reaches higher fidelity, while the symmetric fold trades $\approx 2\times$ rotations per step for the lowest error. Transcribed from the `schedules_table` block of `results/summary.json`.

schedule	order	fidelity	±95%	rotations	rate p
random (1st)	1st	0.8666	0.0161	118	1.97
coefficient (1st)	1st	0.9403	0.0133	118	1.96
clique (1st)	1st	0.9467	0.0107	118	2.02
antithetic (2nd, free)	2nd	0.9589	0.0139	118	4.07
symmetric (2nd)	2nd	0.9976	0.0009	225	4.03

As seen in Fig. 3, fitting the log–log slope of mean infidelity against the step count r, the three first-order schedules cluster at rate 1.96—their curves lie nearly on top of one another, a visual proof of impotence—while the antithetic and symmetric folds achieve rates 4.07 and 4.03, a clean doubling of the convergence order from ≈ 2 to ≈ 4. The antithetic schedule is not only more accurate than every first-order ordering; it preserves the *identical* $L \cdot r$ rotation count at each r, differing only in per-step direction.

Table II makes the matched-cost reading explicit at a reference budget of $r = 6$ steps (118 mean rotations). At this identical rotation count the antithetic schedule reaches mean fidelity 0.9589 against 0.9403 for the best first-order ordering—a reduction of mean infidelity from 0.0597 to 0.0411, a factor of 1.5, *for free.* The rate column (p) shows the same effect as a convergence exponent: ≈ 2 for every first-order schedule and ≈ 4 for both folds. The symmetric fold reaches the highest fidelity but spends $\approx 2\times$ rotations per step to do so, whereas the antithetic fold matches the first-order budget exactly and still improves on it.

C. The matched-cost frontier: rotations to a target fidelity

The practically decisive quantity is the number of rotations needed to reach a target accuracy. Figure 4 plots infidelity against the realized rotation count for every schedule, and the second-order folds dominate the entire high-accuracy region; because the convergence rates differ, the gap *widens* with budget. The crossing points are summarized in Table III: to reach fidelity 0.99 the best first-order ordering needs 315 rotations, the antithetic schedule 236 and the symmetric schedule 225—a $1.40\times$ reduction. We note that, to reach 0.999, *no* first-order schedule succeeds within the swept budget (entry "—"), while the antithetic and symmetric schedules reach it at

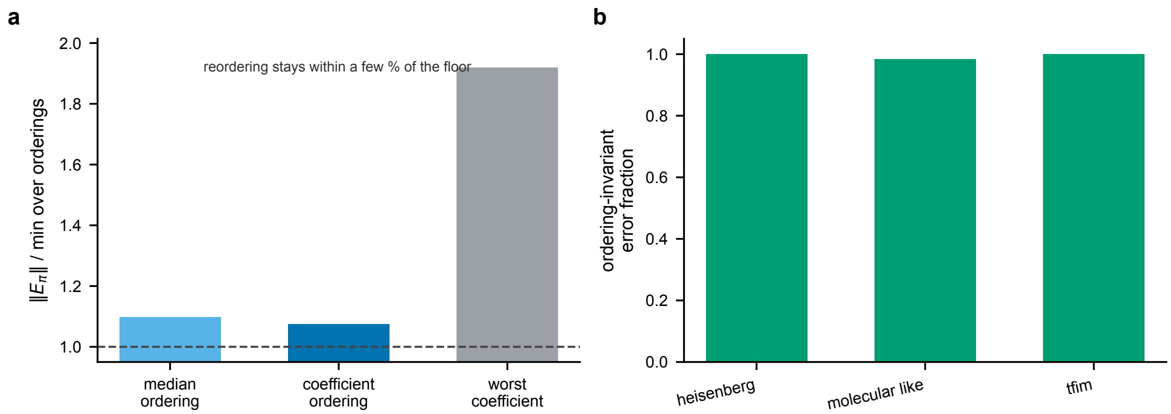

FIG. 2. **The leading error is nearly ordering invariant.** (a) Spectral norm of the leading-error operator E_π for the coefficient ordering and for sampled orderings, divided by the per-instance minimum over orderings; bars are means over the 72 instances with $n \leq 8$ (where the dense spectral norm is tractable). A median ordering is within 1.097× of the floor and the coefficient ordering within 1.074×; the worst single instance reaches 1.92×. The dashed line marks the unreducible floor. (b) The fraction of the leading error that is ordering invariant by family (over all 144 instances): complete on transverse-field Ising and Heisenberg, and 0.98 on the dense molecular-like family. Reordering cannot move what these bars measure.

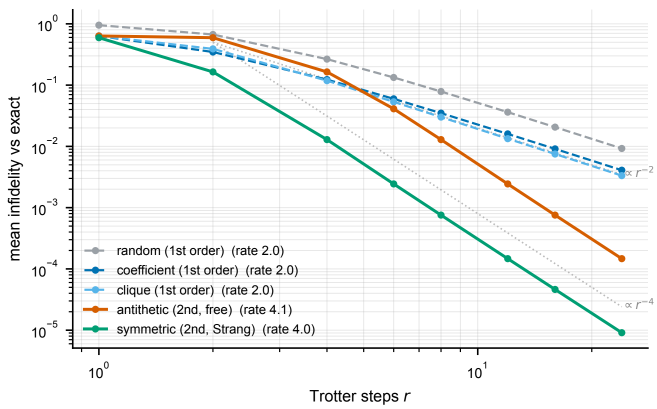

FIG. 3. **The antithetic schedule doubles the convergence order at the same gate count.** Mean infidelity against the exact propagator versus Trotter step count r (log–log), averaged over the 144 instances. The three first-order schedules (cool colors, dashed) lie nearly on top of one another at fitted rate ≈ 1.96 (parallel to the $\propto r^{-2}$ guide), the empirical signature of ordering impotence. The antithetic fold (vermillion), which uses the identical $L \cdot r$ rotations as the first-order schedules at each r, and the symmetric Strang fold (green) track the $\propto r^{-4}$ guide at rates 4.07 and 4.03.

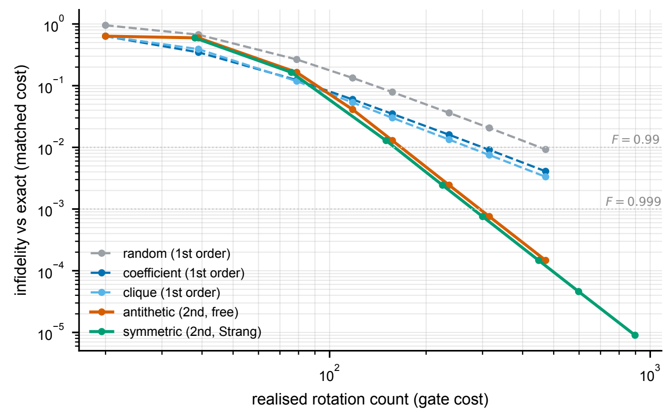

FIG. 4. **Matched-cost frontier: infidelity versus realized rotation count.** Mean infidelity (log) against the realized rotation count (log)—the matched-cost efficiency view—over the 144 instances. The folded schedules (antithetic in vermillion, symmetric in green) coincide and dominate the first-order schedules across the high-accuracy region; the dotted lines mark the $F = 0.99$ and $F = 0.999$ targets. First-order Trotter does not reach $F = 0.999$ within the swept budget; the folds do. The antithetic schedule achieves this at the *same* rotation count as the first-order schedules at each step count.

315 and 300 rotations. Most strikingly, the free antithetic schedule, costing exactly $L \cdot r$ rotations, tracks the 2×-per-step symmetric fold on this matched-cost axis to within line width—the second-order accuracy is recovered at first-order cost.

D. The advantage is size independent

A central question for any product-formula improvement is whether it survives at scale. Because the convergence-order argument of Theorem 2 is size independent, it should; we verify this directly. The matrix-free Krylov reference removes the dense-exponential bottleneck and lets us evaluate the exact benchmark from $n = 4$ up to $n = 14$ qubits, the same range over which a dense $e^{-\mathrm{i}Ht}$ would be intractable. As seen in Fig. 5(a), the fitted convergence rate is flat in n: the three first-order schedules hold a rate ≈ 2 at every size (rate 1.94 at $n = 14$), while the antithetic and symmetric folds hold a rate ≈ 4 (antithetic rate 4.06 at $n = 14$). The doubling

TABLE III. **Rotations to reach a target fidelity (the matched-cost efficiency metric).** For each schedule, the realized rotation count at which the mean fidelity over the 144 instances first reaches 0.9, 0.99, and 0.999; "—" marks a target not reached within the swept step grid. The folded schedules reach 0.99 at a 1.40× smaller budget than the best first-order ordering and are the only schedules to reach 0.999. Transcribed from the `gates_to_target` block of `results/summary.json`.

schedule	$F \geq 0.9$	$F \geq 0.99$	$F \geq 0.999$
random (1st)	157	472	—
coefficient (1st)	118	315	—
clique (1st)	118	315	—
antithetic (2nd, free)	118	236	315
symmetric (2nd)	150	225	300

TABLE IV. **Scaling of the convergence rate and speedup with system size.** For each size n, the fitted convergence rate p of the first-order (1st), antithetic (anti), and symmetric (sym) schedules, the matched-cost rotation speedup to reach $F \geq 0.99$ (best first-order ordering / best fold), and the mean wall-clock time per instance. The first-order rate stays ≈ 2 and the folded rates stay ≈ 4 at every size, while the matrix-free reference keeps the per-instance cost modest up to $n = 14$. Transcribed from the `by_size` block of `results/summary.json`.

n	p (1st)	p (anti)	p (sym)	speedup	time (s)
4	1.98	4.08	4.03	1.68	0.026
6	1.97	4.07	4.02	2.00	0.044
8	1.97	4.06	4.02	1.58	0.070
10	1.94	4.08	4.03	2.09	0.124
12	1.96	4.08	4.03	2.07	0.371
14	1.94	4.06	4.02	2.06	1.835

of the convergence order is therefore not a small-system artifact but a property of the schedule. Figure 5(b) and Table IV show the same conclusion in the matched-cost metric: the free antithetic fold reaches fidelity 0.99 at fewer rotations than the best first-order ordering at *every* size, with the speedup reaching 2.06× at $n = 14$. This scaling is obtained with the antithetic schedule's identical $L \cdot r$ rotation count, so the advantage is recovered at zero overhead independently of system size.

E. Instance dependence and the learned scheduler

The gain is not uniform across families, and CGS is designed to exploit exactly this structure. Figure 6 and Table V decompose the rotations-to-target by family. On the dense *molecular-like* and *transverse-field Ising* families the folded schedules cut the cost to reach 0.99 by roughly a third to a half and are the only schedules to reach 0.999. On *Heisenberg*, by contrast, the commuting-clique structure makes the first-order clique schedule essentially *exact* at a single step (Proposition 2): every clique-based schedule reaches 0.999 at the minimal rotation count, and the symmetric fold's 2× per-step overhead is here wasted. The gate-optimal schedule is therefore genuinely instance dependent—fold the hard instances, and leave the fast-forwardable ones first-order.

TABLE V. **Per-family rotations to a target fidelity.** For each family and schedule, the realized rotation count to reach mean fidelity ≥ 0.99 and ≥ 0.999; "—" marks a target not reached within the swept step grid. The folded schedules are the only ones to reach 0.999 on the molecular-like and transverse-field Ising families; the clique-based first-order schedules already reach it at minimal cost on Heisenberg. Transcribed from the `gates_to_target` block of `results/summary.json`.

family	schedule	$F \geq 0.99$	$F \geq 0.999$
heisenberg	random (1st)	—	—
	coefficient (1st)	24	24
	clique (1st)	24	24
	antithetic (2nd, free)	24	24
	symmetric (2nd)	47	47
molecular_like	random (1st)	432	—
	coefficient (1st)	432	—
	clique (1st)	432	—
	antithetic (2nd, free)	216	432
	symmetric (2nd)	205	409
tfim	random (1st)	408	—
	coefficient (1st)	272	—
	clique (1st)	272	—
	antithetic (2nd, free)	136	272
	symmetric (2nd)	129	257

The learned scheduler of Sec. II F exploits this structure end-to-end. Trained leave-one-instance-out over the ten commutator-graph features, so that no test instance informs its own predictor, it attains 93% accuracy against the per-instance oracle. We note that, on the subset of 92 instances where every fixed schedule reaches the target, the scheduler needs 114.0 rotations on average, against an oracle optimum of 112.4 and the best single fixed schedules' 117.0 (symmetric) and 117.6 (antithetic)—recovering near-oracle efficiency by routing fast-forwardable instances to the cheap first-order schedule and hard instances to a fold. The scheduler is interpretable and listing-order invariant by construction (Proposition 3): it is a selector over named schedules, not a black-box policy.

IV. DISCUSSION

a. Summary of the result. In this manuscript we introduced commutator-graph scheduling (CGS), and the result it establishes has two halves that must be kept together. *Negatively*, a single term ordering is a provably weak lever for first-order Trotterization: reordering only

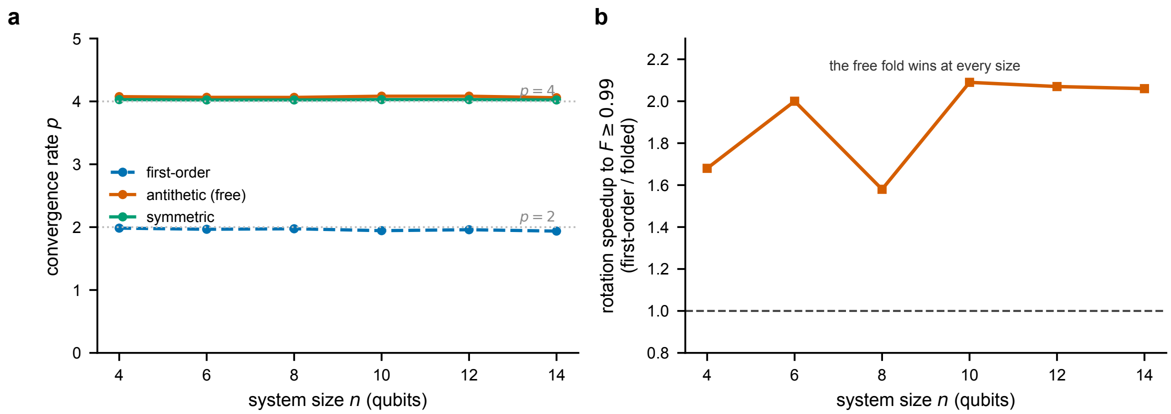

FIG. 5. **The convergence-order doubling and the matched-cost speedup are size independent.** (a) Fitted convergence rate p (mean per-instance log–log slope of infidelity vs. r) versus system size n. The first-order schedule (blue, dashed) holds $p \approx 2$ and the antithetic (vermillion) and symmetric (green) folds hold $p \approx 4$ across the whole range $n = 4$–14; the dotted guides mark $p = 2$ and $p = 4$. (b) Matched-cost rotation speedup to reach $F \geq 0.99$ (best first-order ordering divided by the best fold) versus n; the free fold wins at every size, reaching $2.06\times$ at $n = 14$. The matrix-free Krylov reference, validated against dense `expm` to 10^{-9}, is what makes the exact benchmark tractable to 14 qubits.

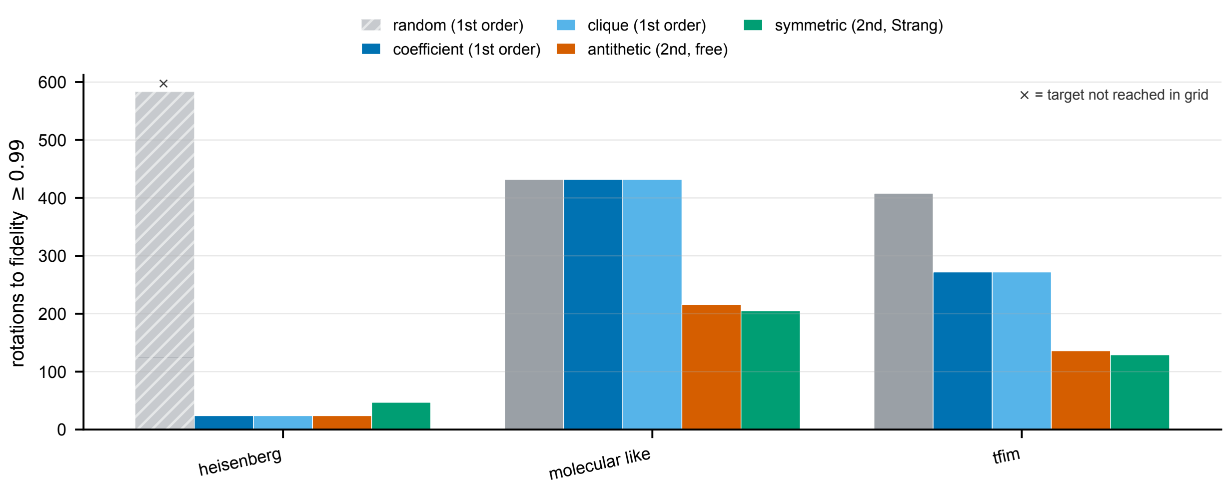

FIG. 6. **Rotations to fidelity** ≥ 0.99 **by Hamiltonian family.** Lower is better; an $\times$ marks a target not reached within the swept budget. On the dense molecular-like and transverse-field Ising families the folded schedules (antithetic, symmetric) reach the target at substantially fewer rotations than any first-order schedule. On Heisenberg the commuting-clique first-order schedules are already near-exact at the minimal budget, so folding only adds overhead. Bars are over the $N = 48$ instances per family.

re-signs the leading-error commutators, the leading-error norm is exactly invariant on collision-free families and within a few percent of its floor otherwise, and no ordering escapes the $O(t^2/r)$ rate—turning the well-known empirical observation that ordering heuristics are within noise into a theorem. *Positively*, the freedom that does matter is the per-step schedule: the antithetic fold cancels the leading error and converges at second order at the identical gate count, reaching a target fidelity with $1.40\times$ fewer rotations and reaching 0.999 where first-order Trotter plateaus. We do not claim a new product-formula *primitive*—the symmetrization mechanism is classical [4, 7]—but rather the framing that the lever is the schedule, that a fixed ordering provably cannot realize it, and that the commutator graph both certifies the impotence and structures the schedule, together with the matched-cost quantification and the learned scheduler.

b. Relation to prior work. The commutator graph and its clique structure are established in measurement grouping for variational algorithms [12–14]; we repurpose them for *time evolution* and, beyond ordering, for scheduling. The sensitivity of Trotterized chemistry to term ordering has been reported empirically [10, 11, 15], and the impotence theorem explains the ceiling those studies encounter. Symmetric (Strang) and higher-order product formulas are classical [4, 7, 27, 28], and randomized compilation—random permutation per step, qDRIFT, stochastic sparsification, randomized multi-product formulas, and composite channels—improves error scaling by averaging rather than cancellation [8, 9, 43, 44, 46]; the antithetic schedule is the *deterministic, zero-variance* counterpart that cancels the same leading term in a single sample, and our matched-cost frontier places it against both the first-order and the randomized baselines. A complementary deterministic route reduces the leading error by taking linear combinations of Trotter circuits—multi-product formulas [45]—at the cost of ancillas or postselection, whereas the antithetic schedule pays no such overhead. Beyond product formulas, query-optimal primitives based on linear combinations of unitaries, truncated Taylor series, quantum signal processing, and qubitization achieve logarithmic precision scaling [29, 32–35], and refined input- and observable-dependent analyses tighten the cost further [47, 51]; these target the fault-tolerant regime, while CGS is a near-term, zero-overhead upgrade. Hardware-aware compilations such as the fermionic-swap network reduce circuit depth for the same Trotter step [50]. The tight commutator-scaling theory of Trotter error [5, 6] is the lens throughout, and the commutator error form is its explicit, matrix-free, ordering-resolved representation.

c. Why a matched-cost schedule matters. For first-order formulas the schedule is genuinely free—it changes no gate, qubit, or depth count—so any accuracy it recovers is pure profit. In the high-accuracy regime where precise chemistry and materials simulation operate, the difference between an $O(t^2/r)$ and an $O(t^3/r^2)$ method at fixed cost is the difference between reaching and not reaching a target, as the unreachable 0.999 column of Table III shows. The commutator graph supplies both halves of the recommendation at no extra cost: it certifies when ordering cannot help, and its clique coloring tells the learned scheduler which instances to fold and which to leave alone.

d. Limitations. The boundaries of the study are explicit. (i) The second-order advantage is a small-step-size effect: at coarse steps the antithetic fold can be *worse* than a good first-order ordering (its curve in Fig. 4 starts above and crosses below near the intermediate budget); we report the crossover. (ii) The matrix-free Krylov reference extends the exact benchmark to $n = 14$ qubits—and we verify there that the advantage is size independent (Sec. III D)—but truly large systems ($n \gtrsim 30$), where even a sparse statevector is intractable and a tensor-network reference is required, are not reached; the exact *spectral-norm* impotence sweep is itself capped at $n \leq 8$ (the matrix-free collision and Hilbert–Schmidt statistics carry Theorem 1 at all sizes). (iii) The families are transverse-field Ising, Heisenberg, and random local two-body molecular-like sums, a proxy for, not an instance of, mapped molecular electronic-structure Hamiltonians at scale [17, 18]. (iv) The cost is the realized rotation count and ignores the gate-synthesis cost of individual rotations and hardware connectivity; the antithetic schedule's seam rotations could be merged to reduce the count further, which we do not exploit so that its budget matches first-order exactly. (v) All results are classical statevector simulations on CPU, with no hardware validation.

e. Outlook. Most critically, because the antithetic upgrade changes no gate, qubit, or depth count, commutator-graph scheduling constitutes a zero-overhead accuracy improvement that any first-order Trotter compilation can adopt unchanged on near-term hardware. Having shown that the advantage is size independent up to $n = 14$ qubits with a matrix-free reference, natural next steps are truly large systems with a tensor-network reference, real mapped molecular Hamiltonians, higher-order folded schedules, and a graph neural policy trained directly on a differentiable matched-cost objective. In short, by recasting the leading Trotter error as a certifiable operator on the commutator graph, CGS reframes term ordering as the weak lever it provably is, and identifies the per-step schedule as the free, structured degree of freedom that simulation should exploit.

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DATA AVAILABILITY

This study uses no external datasets. All Hamiltonians are generated synthetically from fixed random seeds by the accompanying code, and all numerical values, tables, and figures are produced by that code and stored in `results/summary.json`. The code that implements CGS and regenerates every figure, table, and reported value accompanies this manuscript as an installable package (`pip install .`) and will be released in a public repository upon publication.

Appendix A: Numerical methods and protocol

The analytic development—the Pauli algebra, the commutator error form, and the ordering-impotence, antithetic-cancellation, clique-exactness, and listing-order-invariance results—appears with complete proofs in Sec. II. Here we specify the numerical protocol. Three families are generated as $\{(c_j, P_j)\}$ term lists: transverse-field Ising $H = -J\sum_i Z_i Z_{i+1} - h\sum_i X_i$; Heisenberg $H = \sum_i (J_x X_i X_{i+1} + J_y Y_i Y_{i+1} + J_z Z_i Z_{i+1})$; and *molecular-like*, random local two-body Pauli terms with seeded Gaussian coefficients—a tractable stand-in for the dense, irregular anticommutation structure of Jordan–Wigner

/ Bravyi–Kitaev mapped electronic-structure Hamiltonians [17, 18, 49, 52–54]. The reported-scale configuration (`configs/full.yaml`) uses sizes $n \in \{4, 6, 8, 10, 12, 14\}$, eight seeded instances per (family, size), evolution time $t = 1.0$, a reference budget of $r = 6$ steps, and a step grid $\{1, 2, 4, 6, 8, 12, 16, 24\}$ for the convergence and frontier curves, giving 144 instances in total. The exact reference $e^{-\mathrm{i}Ht}|\psi_0\rangle$ is computed either by dense `scipy.linalg.expm` (the ground truth, used for $n \leq 8$) or by the *matrix-free* Krylov reference `expm_multiply` applied to a sparse Hamiltonian assembled term-by-term from one-nonzero-per-column Pauli permutations [Eq. (4) extended to the reference]; the latter never forms a $2^n \times 2^n$ matrix, agrees with dense `expm` to 10^{-9} (the `reference_backend_verified` flag), and is what extends the exact benchmark to $n = 14$. For each instance and schedule we evaluate the product formula, score fidelity and infidelity against the reference, and record the realized rotation count; from the (rotation count, fidelity) curves we read the rotations to each target fidelity and fit the convergence rate. Ordering impotence is measured by sampling 48 orderings per instance and forming the exact leading-error spectral norm on the 72 instances with $n \leq 8$, where the dense $O((2^n)^3)$ spectral norm is tractable; the matrix-free Hilbert–Schmidt surrogate and collision statistics are computed at all sizes. Per-schedule means carry 95% confidence intervals $1.96\,\hat{\sigma}/\sqrt{N}$.

Appendix B: Software, validation, and reproducibility

The pipeline is a CPU-only Python package (`topoham`) depending on NumPy [19], SciPy [20], NetworkX [21], scikit-learn [16], and Matplotlib. Hamiltonians are stored as Pauli term lists (`hamiltonians.py`); the Pauli algebra of Sec. II A—the $O(n)$ parity test (Lemma 1), the cosine–sine rotation kernel (Lemma 2), and the phased-Pauli product—is the inner loop of the exact statevector simulator (`pauli.py`). The commutator graph and matrix-free error form are built in `commutator_graph.py` and `error_form.py`: the greedy coloring partitions terms into commuting cliques (Proposition 2), the error form assembles Eq. (5) as a signed Pauli-string sum over the edges (Definition 1), and the same module computes the ten listing-order-invariant features the learned scheduler reads (Proposition 3). Schedules are realized as flat lists of (term, step-fraction) rotations whose seam-merged length is the gate cost on which all schedules are compared (`schedules.py`, `env.py`); the learned scheduler (`LearnedSchedulePolicy`) is the standardized logistic-regression classifier of Sec. II F, trained leave-one-instance-out. The exact reference is built in `env.py`: `sparse_pauli_matrix` assembles each Pauli string as a one-nonzero-per-column signed permutation directly from index arithmetic in $O(2^n)$, so the sparse Hamiltonian and its `expm_multiply` reference are formed without a dense $2^n \times 2^n$ matrix.

The single source of truth is `results/summary.json`, written by the runner with full provenance (seed, command, platform, package versions, runtime, peak memory) and *three* integrity flags that must pass before any fidelity is produced: `pauli_algebra_verified` (the fast kernel matches dense `expm` to 10^{-9}), `error_form_verified` (the matrix-free error form matches the dense pair-commutator sum to 10^{-9}), and `reference_backend_verified` (the matrix-free Krylov reference matches dense `expm` to 10^{-9} before it is used at scale). On the run reported here (seed 0, Python 3.13, macOS arm64), the experiment completed in ≈ 98 s wall-clock with ≈ 597 MB peak memory, all three flags `true`. The macros, tables, and figures in this manuscript are generated directly from `summary.json`, so no number is entered by hand. The full pipeline is reproduced from the accompanying code by

```
pip install .
make test
export KMP_DUPLICATE_LIB_OK=TRUE
topoham-reproduce \
  --config configs/full.yaml
```

where `make test` checks the Pauli kernel (10^{-9}), the error form (10^{-9}), antithetic second-order convergence, and clique exactness, and `topoham-reproduce` regenerates `summary.json`, the tables, and the figures. All experiments are reproducible on commodity hardware.

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